Name: ______________________

STRAIGHT-LINE GRAPHS, (GRADIENT AND Y-INTERCEPT)

Date: ________________

LO: I can calculate the gradient of a line and identify the y-intercept from a graph or coordinates.

Gradient and y-intercept

The **gradient** measures how steep a line is: rise ÷ run. A positive gradient slopes upward; a negative gradient slopes **downward**.

Y-intercept: The point where the line crosses the y-axis (where x = 0). Written as (0, c).

Quick examples

●	Gradient = $(y_2 - y_1) \div (x_2 - x_1)$	rise over run
●	Points (1, 3) and (4, 9): m = 6 ÷ 3 = 2	change in y over change in x
●	Horizontal line has gradient 0	no rise
●	Gradient = run ÷ rise	it is rise ÷ run
●	Vertical line has gradient 0	it is undefined

Key idea

Gradient = change in y divided by change in x. Pick any two points on the line.

$$m = y_2 - y_1 \div x_2 - x_1$$

Rise over run

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – from two points

Find the gradient of the line through (2, 3) and (5, 9).

Rise = 9 - 3 = 6

Run = 5 - 2 = 3

Gradient = 6 ÷ 3 = 2

m = 2

Subtract y-values for rise, x-values for run.

Example 2 – negative gradient

Find the gradient of the line through (1, 7) and (4, 1).

Rise = 1 - 7 = -6

Run = 4 - 1 = 3

Gradient = -6 ÷ 3 = -2

m = -2

A negative answer means the line slopes downward.

Example 3 – from a graph

A line passes through (0, 4) and (3, 4). Find the gradient.

Rise = 4 - 4 = 0

Run = 3 - 0 = 3

Gradient = 0 ÷ 3 = 0

m = 0

A horizontal line has zero gradient.

Example 4 – finding the y-intercept

A line has gradient 3 and passes through (2, 8). Find the y-intercept.

Use y = mx + c: 8 = 3(2) + c

8 = 6 + c

c = 2, so y-intercept is (0, 2)

y-intercept = 2

Substitute a known point into y = mx + c to find c.

YOUR TURN – PRACTICE (deliberate practice)

1. Gradient from two points

Calculate the gradient.

a) (1, 2) and (3, 8)

b) (0, 5) and (4, 13)

c) (2, 6) and (5, 6)

d) (1, 10) and (3, 4)

2. Positive and negative gradients

Find the gradient and state whether it slopes up or down.

a) (0, 0) and (4, 12)

b) (2, 9) and (5, 3)

c) (-1, 4) and (3, 12)

d) (0, 7) and (7, 0)

3. Fractional gradients

Calculate the gradient. Leave as a fraction if needed.

a) (0, 1) and (4, 3)

b) (2, 5) and (8, 7)

c) (0, 6) and (3, 4)

d) (1, 2) and (5, 5)

4. Finding the y-intercept

Find the y-intercept using $y = mx + c$.

a) m = 2, passes through (3, 10)

b) m = -1, passes through (4, 1)

c) m = 3, passes through (2, 11)

d) m = $\frac{1}{2}$, passes through (6, 7)

5. Mixed gradient problems

Solve each problem.

a) Find m and c for the line through (0, -3) and (2, 1)

b) Line with m = -2 through (1, 5). Find c.

c) Points (3, 4) and (6, 4). Describe the line.

d) Is gradient (4-1)/(2-5) positive or negative?

e) Line through (-2, -3) and (4, 9). Find equation.

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) Gradient = (y2 - y1) / (x2 - x1) ☐

Correct answer: __________

Reason: ________________

b) Gradient = change in x over change in y ☐

Correct answer: __________

Reason: ________________

c) A negative gradient means the line slopes downward left to right ☐

Correct answer: __________

Reason:

d) A vertical line has gradient zero ☐

Correct answer: __________

Reason: ________________

e) The y-intercept is found by substituting x = 0 ☐

Correct answer: __________

Reason: ________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A line passes through **(-1, -5)** and **(3, 7)**. Find the gradient and y-intercept. Where does this line cross the x-axis?

Expression: ________________________

Simplified: ________________________

b) Two points on a line are **(2, k)** and **(6, 3k)**. The gradient of the line is **4**. Find the value of k and the equation of the line.

Expression: ________________________

Simplified: ________________________

TOP TIPS

Read the question carefully. Show all your working step by step. Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!